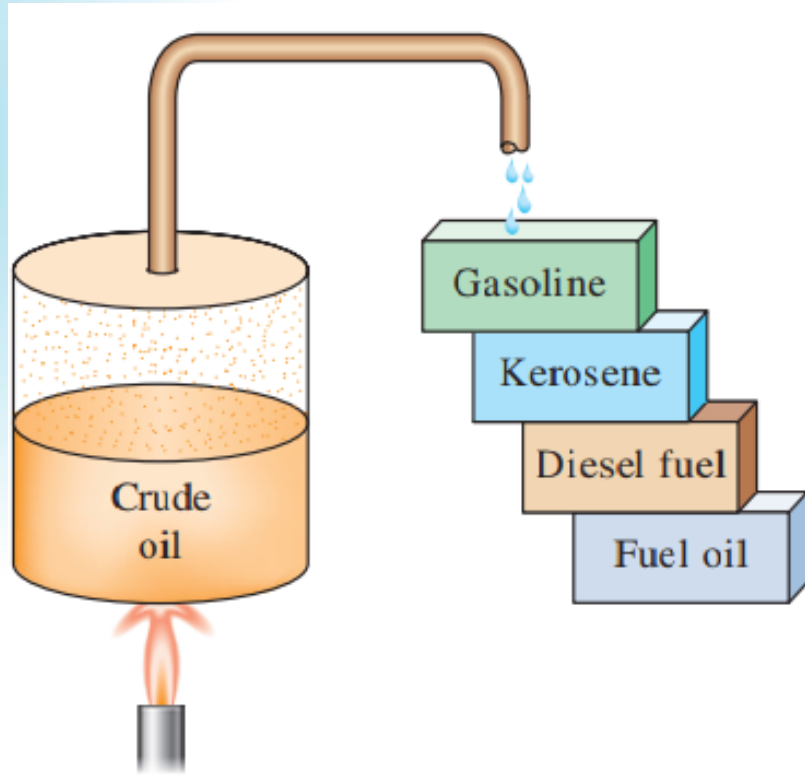


6. Combustion

Introduction

Fuels- The Energy Source

- **A fuel** is any material that can be burned to release thermal energy.
- Most familiar fuels consist primarily of hydrogen and carbon.
- They are called **hydrocarbon fuels** and are denoted by the general formula C_nH_m
- Hydrocarbon fuels exist in all phases, some examples being coal, gasoline, and natural gas.
- The combustible elements in a fuel of any form are carbon, hydrogen, and trace of sulfur.



Most liquid hydrocarbon fuels are obtained from crude oil by distillation.

- Most liquid hydrocarbon fuels are a mixture of numerous hydrocarbons and are obtained from crude oil by distillation.
- The most volatile hydrocarbons vaporize first, forming what we know as gasoline.
- The less volatile fuels obtained during distillation are kerosene, diesel fuel, and fuel oil.

Although liquid hydrocarbon fuels are mixtures of many different hydrocarbons, they are usually considered to be a single hydrocarbon for convenience.

- For example, gasoline is treated as **octane**, C_8H_{18} , and diesel fuel as **dodecane**, $C_{12}H_{26}$.
- Another common liquid hydrocarbon fuel is **methyl alcohol**, CH_3OH , which is also called *methanol* and is used in some gasoline blends.
- The gaseous hydrocarbon fuel natural gas, which is a mixture of methane and smaller amounts of other gases, is often treated as **methane**, CH_4 , for simplicity.
- Natural gas is produced from gas wells or oil wells rich in natural gas.
- It is composed mainly of methane, but it also contains small amounts of ethane, propane, hydrogen, helium, carbon dioxide, nitrogen, hydrogen sulfate, and water vapor.
- On vehicles, it is stored either in the gas phase at pressures of 150 to 250 atm as CNG (compressed natural gas), or in the liquid phase at $-162^\circ C$ as LNG (liquefied natural gas).
- Liquefied petroleum gas (LPG) is a by-product of natural gas processing or crude oil refining. It consists mainly of propane, and thus LPG is usually referred to as propane.
- However, it also contains varying amounts of butane, propylene, and butylenes. Propane is commonly used in fleet vehicles, taxis, school buses, and private cars.

- Ethanol is obtained from corn, grains, and organic waste.
- Methanol is produced mostly from natural gas, but it can also be obtained from coal and biomass.
- Both alcohols are commonly used as additives in oxygenated gasoline and reformulated fuels to reduce air pollution
- Vehicles significantly contribute to air pollutants (nitric oxides, carbon monoxide, hydrocarbons) and the greenhouse gas carbon dioxide. Consequently, the transportation industry is shifting from petroleum-based fuels (gasoline, diesel) to cleaner alternatives like natural gas, alcohols (ethanol, methanol), liquefied petroleum gas (LPG), and hydrogen.

COMBUSTION PROCESS

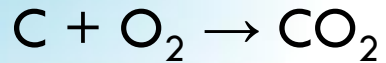
- **Combustion** refers to the rapid oxidation of fuel accompanied by the production of heat, or heat and light. Complete combustion of a fuel is possible only in the presence of an adequate supply of oxygen.
- For combustion to take place, two essential conditions must be satisfied.
 1. Minimum ignition temperatures in atmospheric air are: gasoline 260°C , carbon 400°C , hydrogen 580°C , carbon monoxide 610°C , and methane 630°C .
 2. 3Ts of Combustion

The objective of good combustion is to release all of the heat in the fuel. This is accomplished by controlling the “three T’s” of combustion which are (1) Temperature high enough to ignite and maintain ignition of the fuel, (2) Turbulence or intimate mixing of the fuel and oxygen, and (3) Time sufficient for complete combustion.
- The oxidizer most often used in combustion processes is air, for obvious reasons—it is free and readily available.

- During combustion, nitrogen behaves as an inert gas and does not react with other elements, other than forming a very small amount of nitric oxides.
- Components before and after the reaction are termed **reactants** and **products**, respectively.

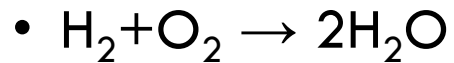
Combustion of individual components

Carbon



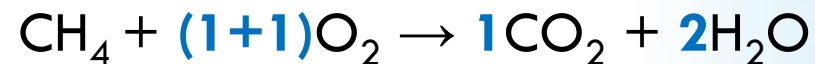
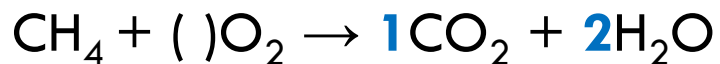
- C and O₂ are the reactants since they exist before combustion, and CO₂ is the product since it exists after combustion

Hydrogen



Example:

Combustion of **methane**: $\text{CH}_4 + ()\text{O}_2 \rightarrow ()\text{CO}_2 + ()\text{H}_2\text{O}$



Balance Carbon (C) and balance (H)

See how much oxygen is needed.

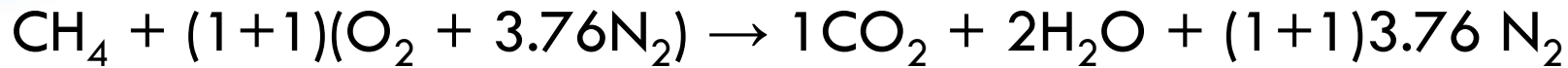
- In combustion analysis, dry air is approximated as 21% oxygen and 79% nitrogen by mole, treating argon as nitrogen and disregarding trace gases.

$$\text{Air} = 0.79/0.21 = 3.76 \text{ kmol of nitrogen/kmol of oxygen}$$

That is,

$$1 \text{ kmol O}_2 + 3.76 \text{ kmol N}_2 = 4.76 \text{ kmol air}$$

Now let's go back to the example of the combustion of methane:

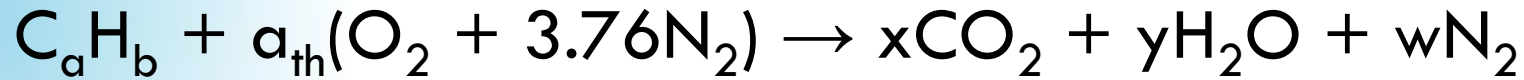


- **$\text{CH}_4 + 2\text{O}_2 + 7.52 \text{ N}_2 \rightarrow 1\text{CO}_2 + 2 \text{H}_2\text{O} + 7.52\text{N}_2$**
- The minimum amount of air needed for the complete combustion of a fuel is the **stoichiometric** or **theoretical air** or **chemically correct amount of air**.
- **Example: $\text{CH}_4 + 2\text{O}_2 + 7.52 \text{ N}_2 \rightarrow 1\text{CO}_2 + 2 \text{H}_2\text{O} + 7.52\text{N}_2$**

- A combustion process is **complete** if all the carbon in the fuel burns to CO_2 , all the hydrogen burns to H_2O , and all the sulfur (if any) burns to SO_2 . That is, all the combustible components of a fuel are burned to completion during a complete combustion process.
- Conversely, the combustion process is **incomplete** if the combustion products contain any unburned fuel or components such as C, H_2 , CO, or OH

General stoichiometric combustion equation:

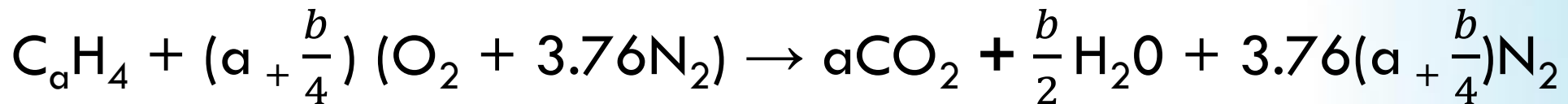
Balance of the combustion equation



Balance

Element	RHS	LHS	Balance
C:	x	a	$a = x$
H:	y	$\frac{b}{2}$	$\frac{b}{2} = y$
O ₂	$x + \frac{y}{2}$	a_{th}	$a_{th} = x + \frac{y}{2} = a + \frac{b}{4}$
N ₂	w	$3.76a_{th}$	$w = 3.76(a + \frac{b}{4})$

Plugging the above into stoichiometric equation,



Air-fuel Ratio (AF)

It is usually expressed on a mass basis and is defined as *the ratio of the mass of air to the mass of fuel* for a combustion process.

That is,

$$\mathbf{AF} = \frac{m_{air}}{m_{fuel}} = \frac{(NM)_{air}}{(NM)_{fuel}} = \frac{(N_{N_2}M_{O_2}) + (N_{O_2}M_{N_2})}{(N_C M_C) + (N_H M_H)}$$

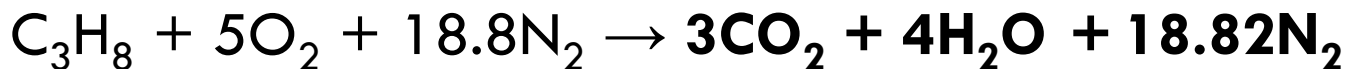
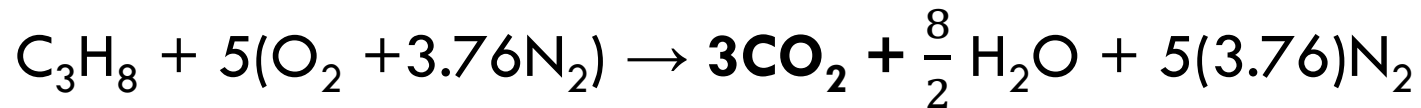
Example:

Write the stoichiometric equation and find the air-fuel ratio for the combustion of propane.

Solution

From Table A-1 the chemical formula of propane is C_3H_8 , Molar mass of propane is 44.1.

$$a + \frac{b}{4} = 5$$



Air-Fuel Ratio (AF)

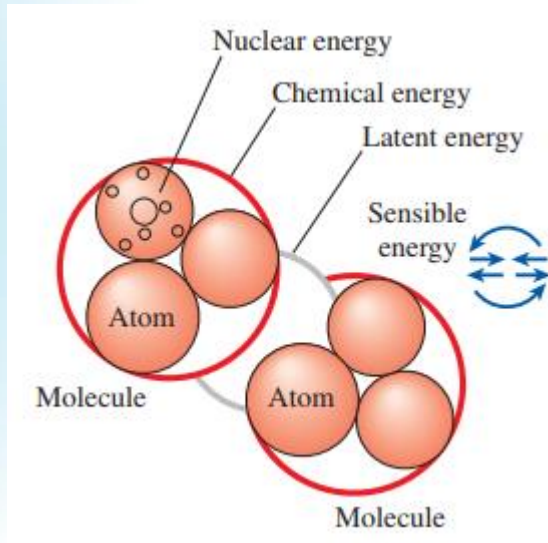
$$AF = \frac{\left(a + \frac{b}{4}\right)(3.76)(M_{air})}{N_{fuel}M_{fuel}} = \frac{5(4.76)(28.97)}{(1)(44.1)} = \mathbf{15.63}$$

- Notice that the products of the theoretical combustion contain no unburned methane and no C, H₂, CO, OH, or free O₂.
- In combustion processes, **excess** air promotes complete burning and controls peak temperatures.
- The amount of excess air is usually expressed in terms of the stoichiometric air as **percent excess air** or **percent theoretical air**.
For example, 50 percent excess air is equivalent to 150 percent theoretical air, and 200 percent excess air is equivalent to 300 percent theoretical air.
- Of course, the stoichiometric air can be expressed as 0 percent excess air or 100 percent theoretical air.
- Amounts of air less than the stoichiometric amount are called **deficiency of air** and are often expressed as **percent deficiency of air**.

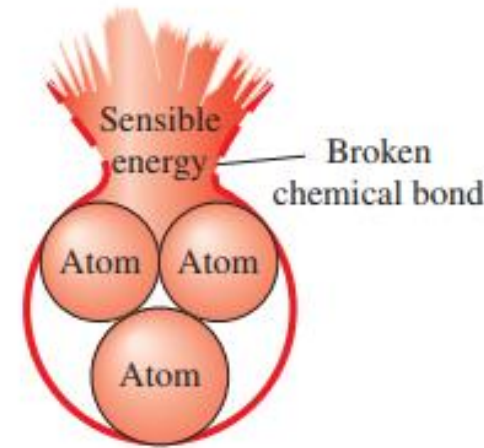
ENTHALPY OF FORMATION AND ENTHALPY OF COMBUSTION

- During a chemical reaction, some chemical bonds that bind the atoms into molecules are broken, and new ones are formed.
- The chemical energy associated with these bonds, in general, is different for the reactants and the products.
- Therefore, a process that involves chemical reactions involves changes in chemical energies, which must be accounted for in an energy balance.

$$\Delta E_{\text{sys}} = \Delta E_{\text{state}} + \Delta E_{\text{chem}}$$

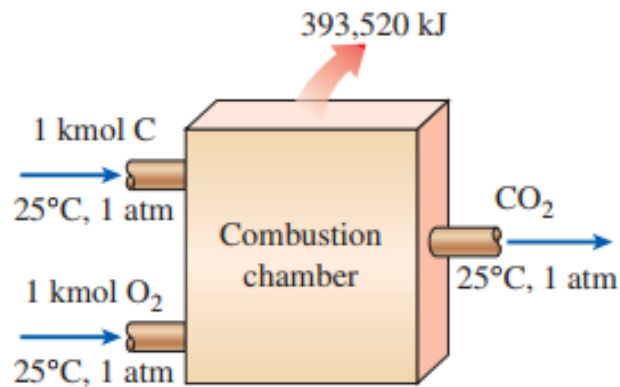


The microscopic form of energy of a substance consists of sensible, latent, chemical, and nuclear energies.

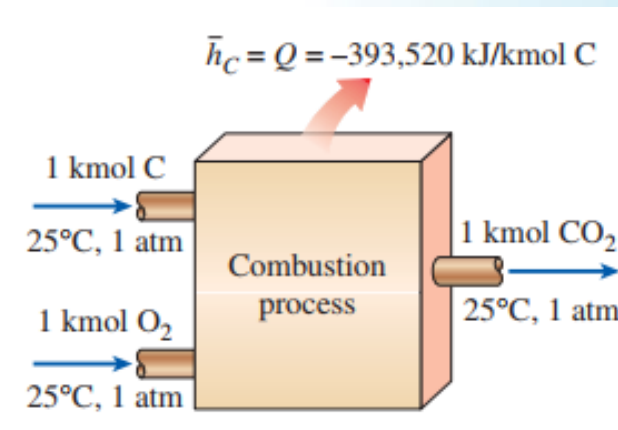


When the existing chemical bonds are destroyed and new ones are formed during a combustion process, usually a large amount of sensible energy is absorbed or released.

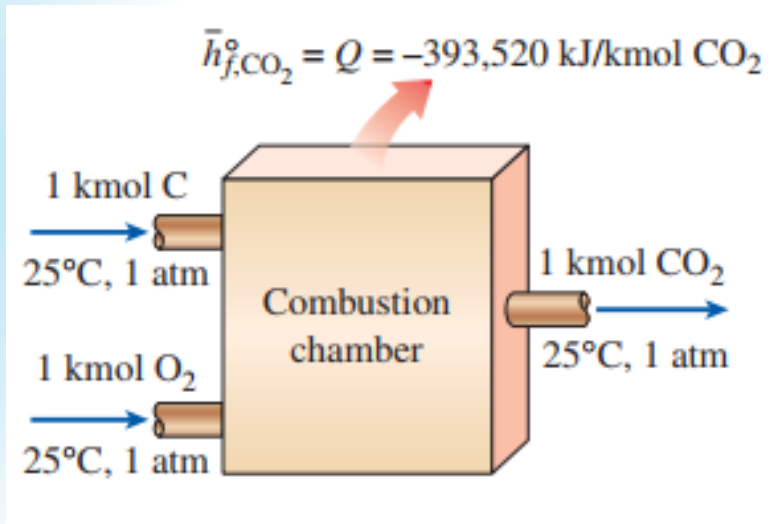
- In this case it becomes necessary to have a common reference state for all substances. The chosen reference state is 25°C (77°F) and 1 atm, which is known as the **standard reference state**.
- Property values at the standard reference state are indicated by a superscript (o) (such as h° and u°).



The formation of CO₂ during a steady-flow combustion process at 25°C and 1 atm.



The enthalpy of combustion represents the amount of energy released as a fuel is burned during a steady-flow process at a specified state.



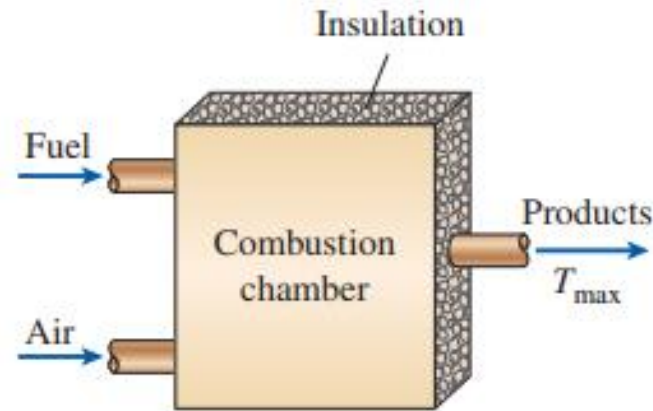
The enthalpy of formation of a compound represents the amount of energy absorbed or released as the component is formed from its stable elements during a steady-flow process at a specified state.

Heating value of the fuel, which is defined as the amount of heat released when a fuel is burned completely in a steady-flow process and the products are returned to the state of the reactants. The heating value depends on the phase of the H₂O in the products. The heating value is called the **higher heating value (HHV)** when the H₂O in the products is in the liquid form, and it is called the **lower heating value (LHV)** when the H₂O in the products is in the vapor form.

ADIABATIC FLAME TEMPERATURE

The smaller the heat loss, the larger the temperature rise.

In the limiting case of no heat loss to the surroundings ($Q = 0$), the temperature of the products reaches a maximum, which is called the **adiabatic flame** or **adiabatic combustion temperature** of the reaction.



The temperature of a combustion chamber becomes maximum when combustion is complete and no heat is lost to the surroundings ($Q = 0$)